

Linear Algebra · Week 3

9 October 2026 · Original practice · Student copy · No solutions

Try independently, compare with a partner, then explain why your method works. Write complete reasons. These are practice activities, not hand-in questions.

Activity 1

For A, B and C, decide whether the determinant is zero without cofactor expansion. Give a specific structural reason. For C, also give its determinant. Do zeros on the diagonal alone force a zero determinant?

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 0 & 0 \\ 1 & 4 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 0 & 1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & -1 & 4 \\ 0 & -3 & 1 \\ 0 & 0 & 5 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 2 \\ 3 & 0 \end{pmatrix}$$

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Activity 2

Find $\det(M)$ using row operations. Record how each operation affects the determinant. Stop as soon as a triangular matrix appears. Explain why your final answer refers to M rather than just the final matrix.

$$M = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 4 & 3 \end{pmatrix}$$

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Activity 3

Calculate $\det(N)$ by cofactor expansion. Choose an efficient row or column and write the sign and minor explicitly. Then check using a different row or column. Explain why the two answers must agree.

$$N = \begin{pmatrix} 2 & 0 & 3 \\ 1 & -2 & 0 \\ 4 & 0 & 1 \end{pmatrix}$$

Activity 4

Without lengthy calculation: (a) Find $\det(T)$. (b) If S is obtained by swapping rows 1 and 3 of T , find $\det(S)$. (c) A further row replacement $R_2 \leftarrow R_2 + 5R_1$ gives V : find $\det(V)$. Write one sentence naming the determinant rule you most need to practise.

$$T = \begin{pmatrix} 4 & 2 & -1 \\ 0 & -1 & 3 \\ 0 & 0 & 2 \end{pmatrix}$$

